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A SVM-based detection method for electricity stealing behavior of charging pile

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Abstract

With the continuous growth of electric vehicles, the electricity stealing behavior of charging pile is becoming more and more frequent. In order to protect the safety of power grid, effective monitoring should be made for electricity stealing behavior. In this paper, a method based on support vector machine (SVM) is proposed to detect electricity stealing behavior of charging piles. By constructing a recognition model of electricity stealing behavior of charging pile, the purpose of anti stealing electricity of charging pile is achieved. In the model feature data input stage, the relevant features of electricity stealing behavior are extracted, and the support vector machine classifier is used for training, and the classification effect is tested on the test set. Experimental results show that the model has high accuracy. It can meet the detection requirements of electricity stealing behavior of charging pile.

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Keywords: Charging pile; electricity stealing monitoring; support vector machine; power grid security

1. Introduction

With the wide use of fossil fuels, environmental pollution and energy crisis are becoming more and more serious, which has become a common problem faced by all mankind. Under the guidance of national policies, the rapid development of electric vehicles has eased the pressure on energy and environment. However, with the continuous growth of electric vehicles, higher requirements are put forward for charging piles. The rapid increase in the number

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of charging piles is a huge challenge to the power supply system of the whole city. In addition, distribution networks with higher automation, better security and reliability are needed [1].

The electricity stealing problem of charging pile has always troubled the power supply department, and the electricity stealing behavior makes the charging process have potential safety hazards. The electricity stealing behavior of electric vehicle charging pile is different from that of ordinary household users, so it is difficult to build a portrait of electricity consumption behavior for users [2]. Due to the strong liquidity of public charging pile, the monitoring of electricity stealing behavior needs to be more accurate and timely, and the electricity stealing behavior may cause different degrees of impact on other charging piles in the site. Facing the liquidity of public charging pile and the concealment of electricity stealing, it needs more effective detection methods. In this paper, through the analysis of the characteristics of electric vehicle and electric vehicle charging pile, a kind of electricity stealing behavior detection of charging pile based on support vector machine is designed. This paper makes a comprehensive analysis of the electricity consumption information of the charging pile and the electricity consumption habits of the users, so as to determine whether there is electricity stealing behavior and make corresponding measures in time.

2. Related research

Aiming at the current common power stealing technology and the defects of power grid equipment, large-scale improvement has been carried out, such as 5G communication, intelligent video real-time monitoring, control terminal monitoring mode [3]. Electric vehicle and charging pile electricity information acquisition system is gradually building and improving. It has accumulated more data, and researchers study the electricity stealing behavior of charging pile from the perspective of data mining.

Reference [4] used the method of logical regression to analyze and detect the electricity consumption behavior combined with the characteristics of the charging pile, but this method involves a lot of iterative operations, which is not a small cost. In reference [5], various characteristics of electricity consumption behavior were extracted, and the existence of electricity stealing behavior was evaluated by setting a threshold and then weighting the results. Reference [6] uses clustering algorithm to model the characteristics of electricity consumption behavior, but the complexity of charging behavior makes the overall monitoring effect of the model need to be improved. Reference [7-8] used neural network to detect the electricity stealing behavior of charging pile. Deep neural network can automatically extract and select features with low requirements, but it needs a large amount of data. At the same time, it requires high calculation requirements, and the training process is very time-consuming. Reference [9] considered the abnormal monitoring of electric vehicle charging pile. This paper proposes an unsupervised abnormal power consumption mode detection, analyzes the characteristics of the electricity consumption behavior of the charging pile, and uses the principal component analysis method to reduce the dimension of the data, and detects whether there is electricity stealing behavior according to the abnormal electricity consumption of the charging pile. Reference [10] considers that electric vehicle users have accumulated a large number of user data on the charging pile. The user's characteristics of charging point location, charging time and charging power consumption are analyzed. It builds a model based on electricity consumption for users of electric vehicle public charging pile, but there is a lag in the acquisition of user data. To sum up, although many algorithms are applied to the monitoring of electric vehicle charging pile's electricity stealing behavior. However, there is still a lack of a hybrid method for private and public charging points, and the accuracy of monitoring needs to be further improved. In this paper, a method based on support vector machine (SVM) is proposed to detect electricity stealing behavior of charging piles. Methods by constructing an efficient and feasible identification model of electricity stealing behavior of charging pile, the purpose of anti stealing power of charging pile was achieved. In the stage of model feature data input, the relevant data and abnormal data of electricity stealing behavior are effectively extracted and analyzed. Then the SVM classifier is trained by the training sample set, and the test sample set is used to detect whether there is electricity stealing suspicion. Finally, the classification results are compared with the real situation.

3. The whole model of electricity stealing behavior identification method for charging pile

The identification method of electric larceny behavior of charging pile achieves the purpose of anti stealing power of charging pile by constructing an efficient and feasible identification model of electricity stealing behavior of charging pile. In the feature data input stage of the model, the relevant data and abnormal data of electricity stealing behavior are effectively extracted and analyzed. After preprocessing the sample data, a diagnosis model of electric vehicle charging pile users' electricity stealing behavior is established by using support vector mechanism. The recognition of stealing electricity is realized by using sample feature data as input and output sample classification. A recognition model of charging pile stealing electricity is built, and then the model is used to realize real-time diagnosis. The key to identify electricity stealing behavior of electric vehicle charging pile through data mining is to extract electrical feature information, user information feature and power time sequence feature under electricity stealing condition [11-12]. The overall method flow steps are shown in Figure 1.

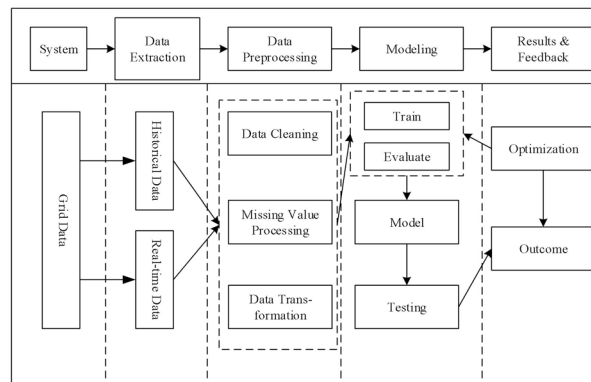


Fig.1. Model flow chart

There are five main steps in the whole process

1. The sample data collected by electric vehicle charging point includes a series of data such as electricity stealing samples, normal samples without stealing electricity, user files, etc.
2. All the collected data are classified according to historical data and real-time data, so as to facilitate the later training and detection.
3. The collected data are analyzed and preprocessed.
4. In the stage of model identification, this paper uses the method of detecting electricity stealing behavior of charging pile based on support vector machine. The whole model is divided into two steps: firstly, the historical sample data is used for modeling, and then a discrimination model is constructed; secondly, the pre-processing real-time data is used to judge whether the current electric vehicle charging pile users have electricity stealing behavior.
5. Real time data is constantly changing. After verifying the real-time data in a certain period of time, it is classified as historical data.

4. Algorithm analysis of support vector machine

4.1. Basic idea of SVM

The support vector machine model can be described as follows:

$$f(x) = \omega^T x + b \quad (1)$$

In fact, there are many lines that can separate the two symbols. As shown in the figure, both lines completely separate the two symbols. Although the dotted line can also separate the two symbols, it lacks generalization. The generalization of solid line is better than that of virtual color. Then, the basic idea of SVM can be briefly summarized as to find the best line, and the greater the distance from the sample point, the better.

4.2. Establishment of SVM model

The dataset can be described as $D = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^p$, $y_i \in \{1, -1\}$. The purpose of support vector machine is to divide the distance between these points as large as possible and find a hyperplane to separate them. The model expression is as follows. Since y_i and $\omega_i^T x + b$ are the same sign, the model is rewritten as follows:

$$\begin{aligned} & \text{Max margin } (\omega, b) \\ \text{s.t. } & y_i \cdot (\omega_i^T x + b) > 0 \quad (i = 1, 2, \dots, N) \end{aligned} \quad (2)$$

Mark a straight line on the plane as: $y = \omega^T x + b$, and the distance from point (x_i, y_i) to the line can be expressed as: $\text{distance} = \frac{1}{\|\omega\|} |\omega^T x + b|$.

Select the hyperplane and make these points as far away from the hyperplane as possible. Then the point closest to the hyperplane can be written as $\min \text{distance}(w, b, x_i)$. The distance needs to be separated as far as possible. The model is further rewritten as follows:

$$\begin{aligned} & \max_{\omega, b} \min_{x_i} \frac{1}{\|\omega\|} |\omega^T x + b| \\ \text{s.t. } & y_i \cdot (\omega_i^T x + b) > 0 \quad (i = 1, 2, \dots, N) \end{aligned} \quad (3)$$

ω and b can be freely retracted and retracted. In order to facilitate the calculation, the model is further simplified, and to minimize the problem:

$$\begin{aligned} & \min_{\omega, b} \frac{1}{2} \omega^T \omega \\ \text{s.t. } & y_i \cdot (\omega_i^T x + b) \geq 1 \quad (i = 1, 2, \dots, N) \end{aligned} \quad (4)$$

This is a convex optimization problem with n constraints.

4.3. Model solving

In order to solve the problem easily, the model is rewritten to solve the constrained extremum. Lagrange number multiplication is used. Lagrange function is defined as:

$$L(\omega, b, \lambda) = \frac{1}{2} \omega^T \omega + \sum_{i=1}^N \lambda_i (1 - y_i \cdot (\omega_i^T x_i + b)) \quad (5)$$

The goal of the model is $\min_{\omega, b} L(\omega, b, \lambda)$. Calculate the derivative of ω, b and make it be 0, and find its parameter. The partial derivative of b is:

$$-\sum_{i=1}^N \lambda_i y_i = 0 \quad (6)$$

The partial derivative of ω is:

$$\omega - \sum_{i=1}^N \lambda_i y_i x_i = 0 \quad (7)$$

ω^* has been found. However, the model does not get the equation related to b when seeking $\frac{\partial L}{\partial b}$. Support vector (x_k, y_k) is used to make $1 - y_k \cdot (\omega^T x_k + b) = 0$. Change formula to obtain b^* :

$$b^* = y_k - \omega^T x_k = y_k - \sum_{i=1}^N \lambda_i y_i x_i^T x_k \quad (8)$$

The whole SVM model has been established and solved.

The model is $f(x) = \text{sign}(\omega^* x + b^*)$ and the hyperplane is $\omega^* x + b^* = 0$. Where $\omega^* = \sum_{i=1}^N \lambda_i y_i x_i$, $b^* = y_k - \sum_{i=1}^N \lambda_i y_i x_i^T x_k$.

5. Feature recognition method of stealing electricity

The identification method of electricity stealing behavior in this paper mainly includes two parts:

- Based on multi feature recognition of electric vehicle charging pile users' abnormal electricity consumption. In this paper, the historical sample data is used for modeling, and a discriminant model is constructed to recognize the abnormal data of electricity stealing.
- Based on the basic data and power consumption characteristic data of charging pile users, the real-time data after preprocessing is discriminated. SVM algorithm is used to judge whether there is electricity stealing behavior of users and complete the identification of electricity stealing behavior.

According to the characteristics of electricity stealing and the measurement index affected by electricity stealing behavior. The model features are extracted from instantaneous voltage, instantaneous current, phase imbalance, daily power consumption and monthly maximum demand. It mainly covers the electricity consumption information of charging pile and the electricity consumption habits of specific users.

The main features are as follows:

- Time series of electricity consumption: total daily electricity consumption, daily average electricity consumption, peak power consumption and valley power consumption.
- Average meter reading: statistics the reading of single charging pile as the input of power consumption base.
- Monthly maximum demand: during the observation period, the maximum power demand of charging pile in each month.
- Line loss of the station area: the line loss rate, number of users and power consumption of the user area reflect the degree of line loss and the distribution proportion of power stealing users.
- Power factor: whether there is abnormal reduction of power factor.
- User information: power consumption category, historical electricity stealing number, single charging time, interval time of each charging, payment record, etc.

The diagnosis model of electricity stealing behavior of electric vehicle users is the core content of this method, and the data processing principle is support vector machine algorithm. It mainly includes two parts: model training and model testing. The specific model construction process is shown in Figure 2.

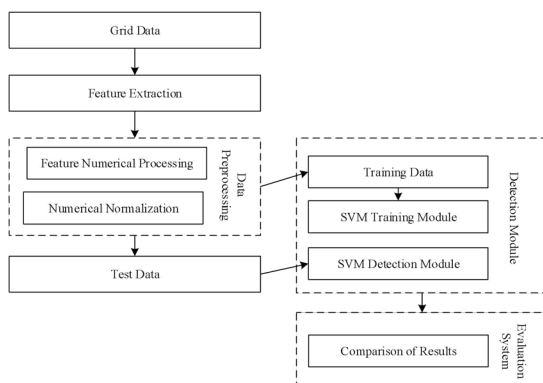


Fig. 2. Algorithm detection flow

It consists of five steps:

- The original sample data is preprocessed to avoid data omission and eliminate significant outliers.
- For the extracted feature quantity, the training sample set and the test sample set are divided.
- The SVM classifier is trained by training sample set.
- The test sample set is used to detect whether there is suspicion of stealing electricity.
- The classification results are compared with the real situation, and the accuracy and recall rate of the analysis model are used as evaluation indexes.

6. Experimental verification

In order to verify the feasibility and effectiveness of this method, this paper is based on the charging equipment electricity information acquisition system in some areas of Beijing as the background implementation platform, and as the basic data source.

6.1. Experimental result

In this paper, k-nearest neighbor model, logistic regression model and support vector machine model with satisfactory accuracy are selected. The data of charging piles collected in some areas of Beijing are tested. The accuracy and recall rate of the final test results are shown in Table 1.

Table 1. Algorithm accuracy table

Test Item	KNN	Logistic regression	SVM
Accuracy	94.23	95.33	98.84
Recall	97.34	93.96	96.72

In addition, due to the concealment of electricity stealing behavior and the lag of data return, it is sometimes necessary to distinguish whether there is electricity stealing behavior in the initial stage of power stealing with few data samples. Therefore, this paper also designs experiments to study the number of training samples, and the results are shown in Figure 3.

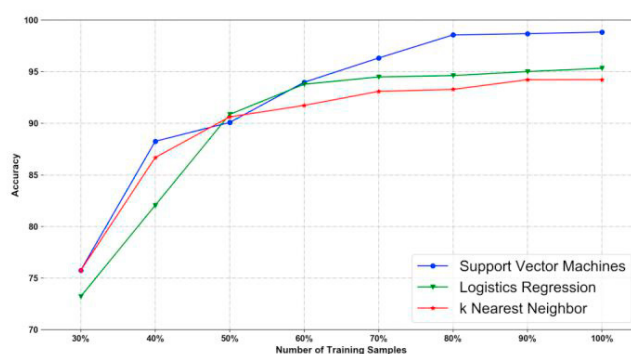


Fig. 3. Comparison of detection accuracy

The detection accuracy rate of electricity stealing behavior in Figure 4 can fully verify that the classification algorithm based on support vector machine can still achieve more than 95% detection accuracy under the condition of limited training samples. When the training sample reaches 80%, the classification accuracy reaches 98.5%. Therefore, it can be considered that sufficient training samples are helpful to improve the detection accuracy, but when the training samples are good, it also has a good performance of electricity theft detection.

The influence of different SVM and different kernel functions on classification results is considered. In this paper, we do not use the performance of kernel function in 40% and 80% of the data volume. Generally speaking, Gaussian kernel has the best accuracy. The accuracy under different data volumes is shown in Figure 4.

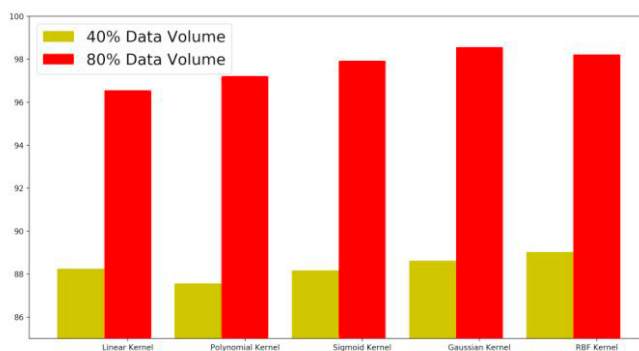


Fig. 4. Accuracy under different Gaussian kernels

7. Conclusion

This paper proposes a detection model of electricity stealing behavior of charging pile based on support vector machine algorithm, which extracts features from the data of electric vehicle charging pile such as electric quantity, voltage, current and power factor. On this basis, SVM algorithm is used to classify electricity stealing behavior. And make full use of the advantages of SVM classification performance. The accuracy and recall rate of this algorithm are 98.56% and 96.72% respectively, which are better than other common classifiers. The effectiveness of the proposed feature extraction method and support vector machine algorithm in detecting electricity stealing behavior of charging pile is verified. The follow-up work can further excavate the association between power consumption time series characteristics and electricity stealing, strengthen the features, improve the classification accuracy, recall rate and other indicators, and further improve the electricity theft behavior detection model.

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